

WASP-140b

R-2276

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-140_201021 · created 2026-08-16T23:30:15+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459143.861 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1719 +/- 0.0088
Transit depth (Rp/Rs)^2	2.96 +/- 0.3 [%]
Orbital Inclination (inc)	83.67 +/- 0.2
Airmass coefficient 1 (a1)	1.3334 +/- 0.0083
Airmass coefficient 2 (a2)	-0.173 +/- 0.011
Scatter in the residuals of the lightcurve fit is	0.62 %
Best Comparison Star	None
Optimal Aperture	5.47
Optimal Annulus	16.23
Transit Duration (day)	0.0702 +/- 0.0031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1719 ± 0.0088 vs 0.1432 (deviation 3.26σ)
Duration fit vs published	0.0702 d vs 0.0604 d (deviation 3.16σ)
Mid-time O–C	-4.8 min
Depth SNR	19.5
Residual scatter	0.62 %

Light curve

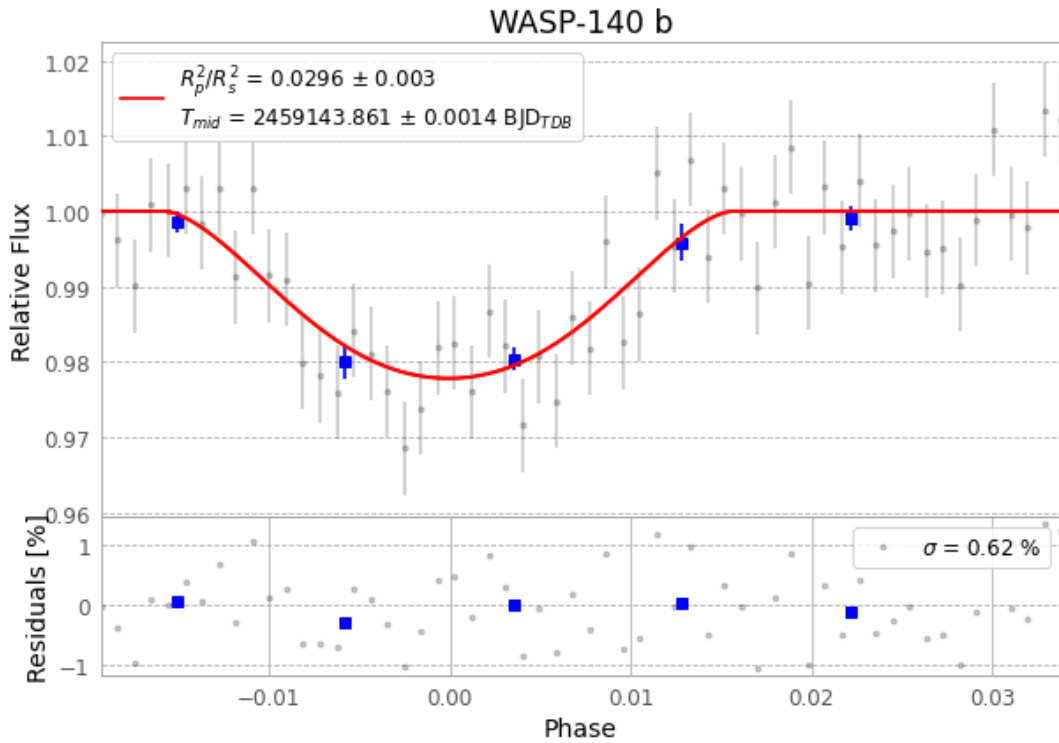


Figure 1 — FinalLightCurve_WASP-140 b_2020-10-21.png (SHA-256 6272d38c6f029960...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [81, 229], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3cd504c0ca994106...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

58 FITS frames (38 MB), WASP-140201021073012.FITS → WASP-140201021102112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-140-201021.log (fd8a7d55b77f...), FinalLightCurve_WASP-140 b_2020-10-21.png (6272d38c6f02...), FinalLightCurve_WASP-140 b_2020-10-21.csv (0723915bc7ad...)

Compute resources

Wall time 107.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 23:30Z.